

8 Zinc

Zinc supports enzymes, growth regulation, leaf expansion, and shoot development. High pH, bicarbonate, excess phosphorus, and weak roots commonly reduce effective uptake.

Why it matters

- **Growth regulation.** Zinc is involved in enzyme activity, hormone processes, leaf expansion, internode growth, and fruit development.
- **Low mobility.** Zinc does not move freely in soil or through the plant, so timing, placement, and tissue coverage matter.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists about 1 to 20 ppm as deficient, 30 to 150 ppm as adequate, and more than 300 ppm as excess.
Sampling caution	Recent zinc sprays and surface residues can make a leaf analysis look higher than internal plant status.

Main soil and environmental controls

Factor	What it can mean in the root zone
pH and bicarbonate	High pH and carbonate-rich conditions reduce soluble Zn and limit uptake.
Phosphorus interaction	High P can worsen Zn deficiency through soil and plant processes. A P input should not be judged without Zn context.
Organic matter and microbes	Organic ligands may either hold Zn in solution or bind it, depending on the compound, pH, and decomposition conditions.
Texture and placement	Zn is easily fixed near the application point in many soils. Soil or fertigation placement should reach active roots without creating a concentrated salt zone.
Water and roots	Dry, cold, saturated, saline, or diseased root zones restrict uptake and root exploration.

Field clues and checks

- **Recognize the pattern.** Small narrow mottled leaves, shortened internodes or rosetting, weak shoots, and small round fruit can occur with serious Zn shortage.
- **Review P and pH.** Check phosphorus history, soil and water pH, bicarbonate, salinity, and feeder roots.
- **Verify foliar response.** A foliar treatment mainly reaches sprayed tissue and may move little to later growth; judge the next flush and compare treated and untreated areas.

DST interpretation Phase 1 placed Zn within a broader nutrient-state pattern and treated Zn by B as a research candidate. Use it to prioritize measurements, not to infer that adding Zn or B will raise yield.